Cryptographic Hashing

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Hashing

Day 1: explain git.

Day 2: the rest

What is Hashing?

A cryptographic hash function is a one way mathematical function that maps data of an arbitrary size to a bit array of fixed size. The output bit array is commonly called a hash or digest. -Wikipedia

Key tenets of a hash algorithm:

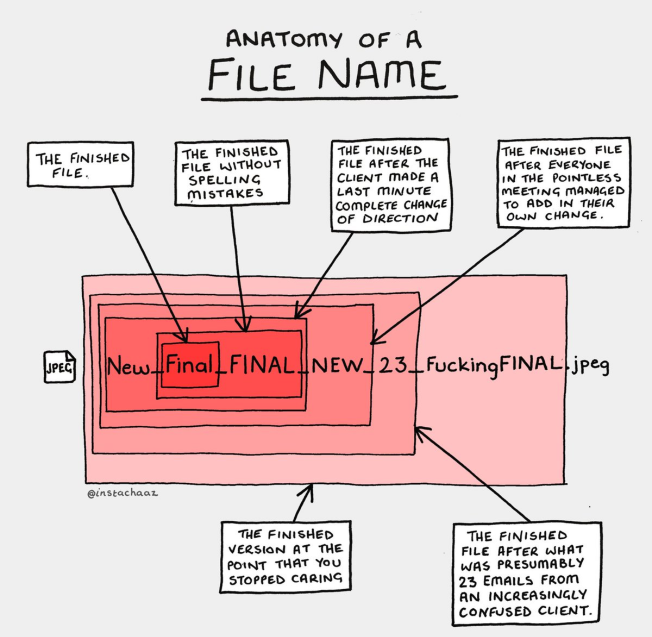
1. ***Fast***, must not take a lot of resources or time to compute (we will violate this rule later...)
2. ***Irreversible***, must not be able to retrieve the original data from the resulting hash (one way transformation)
3. ***Deterministic***, for the same input value, a hash function must always provide the same output
4. A small change to the input should result in large change to the hash value (***avalanche effect***)

In English

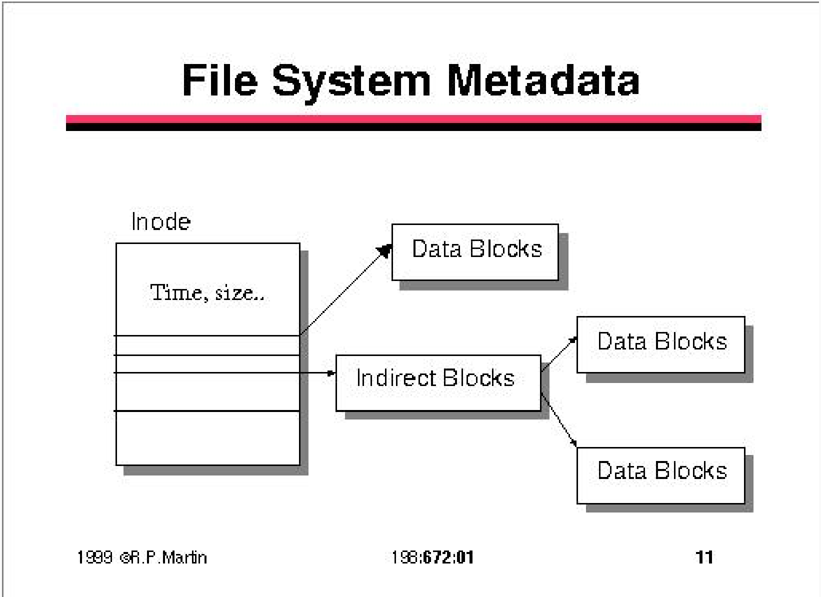
A fancy algorithm that converts a variable length input data to a fixed length output data with the following constraints:

- same input $==$ same output
- different input $==$ ***MASSIVE*** likelihood of different output
- output data reveals ***NO*** information about the input data
- must be **collision resistant**

File names


ANATOMY OF A
FILE NAME
THE FINISHED FILE.
THE FINISHED FILE WITHOUT SPELLING MISTAKES
THE FINISHED FILE AFTER THE CLIENT MADE A LAST MINUTE COMPLETE CHANGE OF DIRECTION
THE FINISHED FILE AFTER EVERYONE IN THE POINTLESS MEETING MANAGED TO ADD IN THEIR OWN CHANGE.
JPEG
New_Final_FINAL_NEW_23_FuckingFINAL.jpeg
@instachaaz
THE FINISHED VERSION AT THE POINT THAT YOU STOPPED CARING
THE FINISHED FILE AFTER WHAT WAS PRESUMABLY 23 EMAILS FROM AN INCREASINGLY CONFUSED CLIENT.

Data (again)


File System Metadata
Inode
Time, size..
Data Blocks
Indirect Blocks
Data Blocks
Data Blocks
1999 ©R.P.Martin
198:672:01
11

Again

- same input $==$ same output
- different input $==$ ***MASSIVE*** likelihood of different output
- output data reveals ***NO*** information about the input data
- must be **collision resistant**

What is a "collision"?

Ship

Figure 3: Ship

Plane

Hashing algorithms

Hashing Algorithm	Hash size (bits)	Hash size (hex length)	Collision-resistant
MD5	128	32	no
sha256	256	64	yes
sha512	512	128	yes
NTLM	128	32	no

Note: "Collision-resistant" should be "no known collisions currently" in the above chart

What is Hashing used for

- verify the integrity of data (block chain, downloads, git commits, secure software)
- digital signatures
- as a part of verifying a given users *Authenticity* (a part of *Authentication*)
- Proof-o'-work (mining, defense against Denial of Service)

Follow along (in `bash`) for some data science fun

What is the difference between the following:

- `echo "Hi!"`
- `printf "Hi!"` <— Hint: Use this one for the exercise

What do the following do?

- | The pipe character
- `awk -F ',' '{ print $1 }'` Hint: pipe the cat output of quiz-data.csv into this
- `grep <string>`

Now about the quiz data, can we reverse our hash?

What are some common (semi) unique identifiers for people?

- campus W-number
- campus UID
- Social Security Number
- First initial Last name combo i.e. `MKijowski`

Day 2 Hashing

Don't believe everything you read on the internet.

Attacks on hashes

- Brute force: hash everything (worst case scenario)
- Pass the hash: an authenticating system accepts hashes, and you have them
- Dictionary attack: you have a dictionary of likely input data used to compute hashes
- Rainbow table: you have a dictionary of pre-computed hashes and known input data
- Collision: you guess/compute ***different*** data that computes to the same hash value

Nonce and salt

Both added to data prior to the hashing function to increase uniqueness. Used for different reasons though.

- Salts increase complexity and prevent several known attacks on hash values (dictionary and rainbow table attacks)
- Nonces are unique (number used only once) and are used to prevent replay attacks (cannot use same nonce) and in proof of work
- These do not increase any guarantee of integrity!!

Key stretching

Suppose an attacker has the salts and hash values and can guess at the original data used...

To make things harder for an attacker, we can simply apply the hashing algorithm to the output of the first.

Do this multiple times to *stretch* (lengthen) the time it takes to attack the hash using common dictionary attacks.

Proof of Work

Proof of an amount of work prior to participating.

Created to reduce email spam.

Example: include a nonce that hashes to X number of leading 0's (see Hashcash)

Lets talk lab 1

- 10 commits == 10 points
- style == 10 points
- task 1 == 50 points (15 points for salted quiz data set)
- task 2 == 50 points (15 points for coins)

Lab 1 continued

- `data/` : folder containing data for use in this lab
- `miner/` : folder for your mining code
- `LAB1-INSTRUCTIONS.md` : markdown file containing lab instructions
- `README.md` : markdown file for your answers and lab writeup. This is the file I am grading (as well as other requested files)
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Markdown Style

- Answers/ responses should be on a new line and not require scrolling around the page (be careful of code blocks).
- Check the style guide in the course repository for more details!!

Recap

Hashing gives you: fixed-size, deterministic, one-way fingerprints of data — with the avalanche effect making them collision-resistant *(for now)*.

- **Integrity & auth** — verify data hasn't changed, prove you know a secret
- **Attacks** — brute force, dictionary, rainbow table, pass-the-hash, collision
- **Defenses** — salt (kills precomputed tables) + nonce (kills replay) + stretching (slows brute force)
- **Passwords & files** — fast hash for integrity, *slow* hash (`bcrypt`/`scrypt`/`Argon2`) for secrets

Now go build a miner and don't leak your coins.

Anatomy of a git commit

Homework

- Research some effective and ineffective password strategies (google it. . .)
- Make a new Linux/WSL user with a weak password (please do not use one of the top 10 passwords)
- Copy the above users weak password hash (their corresponding line in `/etc/shadow`) and submit to Pilot.
    - It should look something like this: `mynewuser2:$y$j9T$hBy.nN91qFxXxCMP...`
- Get started on your lab!!!